

Title:

Cybersecurity Attack Simulation and Incident Response Using ELK Stack

1. Introduction

This project demonstrates a simulated cyberattack environment designed to analyze, detect, and mitigate security threats. The lab environment consists of an attacker machine (Kali Linux) and a target server hosting vulnerable services, including Damn Vulnerable Web Application and a honeypot.

The objective is to:

- simulate real-world attacks
- monitor malicious activity
- implement defensive controls
- analyze logs using a SIEM system

2. Lab Architecture

Components:

- **Attacker Machine:** Kali Linux
- **Target Server:** Ubuntu Server
- **Web Application:** Damn Vulnerable Web Application
- **Honeypot:** Cowrie
- **Web Server:** Apache HTTP Server
- **SIEM:** ELK Stack (Elasticsearch, Logstash, Kibana)
- **Log Shipper:** Filebeat
- **Firewall:** UFW
- **WAF:** ModSecurity

3. Attack Simulation

3.1 Network Reconnaissance (Nmap)

An Nmap scan was performed from Kali Linux to identify open ports and services on the target server.

Observation:

- Port 80 (HTTP) open
- Port 2222 (Cowrie SSH honeypot) open
- Other ports filtered after firewall configuration

3.2 SSH Brute Force Attack (Metasploit)

Using Metasploit Framework, a brute-force attack was executed against port 2222.

Result:

- Connection attempts logged by Cowrie

- Multiple failed login attempts recorded
- No real system compromise occurred

3.3 Web Application Attacks (DVWA)

a. SQL Injection

Payload used:

1' OR '1'='1

Result:

- Authentication bypass achieved
- Database queries manipulated

b. Cross-Site Scripting (XSS)

Payload used:

<script>alert(1)</script>

Result:

- JavaScript executed in browser
- Demonstrated client-side vulnerability

4. Detection and Monitoring

All attack activities were monitored using the ELK Stack.

Log Sources:

- Cowrie logs (SSH attacks)
- Apache logs (web traffic)
- DVWA interactions

Observations:

- Nmap scans generated connection logs
- Metasploit brute-force attempts captured
- SQLi and XSS payloads recorded in HTTP logs

5. Mitigation and Defense

5.1 Firewall Configuration

UFW was configured to reduce the attack surface.

Rules applied:

- Allow port 80 (web traffic)
- Allow port 2222 (honeypot)
- Deny unused ports (e.g., 21, 23, optional 22)

Impact:

- Reduced exposure to scanning
- Limited unauthorized access

5.2 Web Application Firewall

ModSecurity with OWASP CRS was deployed.

Function:

- Inspects HTTP traffic
- Detects malicious payloads

Result:

- SQL Injection attempts flagged/blocked
- XSS payloads detected

6. Incident Analysis

The simulated attacks revealed:

- Open services are primary entry points
- Weak input validation leads to SQLi and XSS
- SSH services are frequent brute-force targets

The honeypot successfully captured attacker behavior, while ELK provided visibility into all activities.

7. Lessons Learned

- Defense-in-depth is essential
- Firewalls reduce exposure but do not stop application attacks
- WAF is necessary for web-based threats
- Continuous monitoring is critical

8. Conclusion

This project successfully demonstrated a full cybersecurity workflow:

- Attack simulation
- Detection using SIEM
- Mitigation using firewall and WAF

The integration of monitoring tools and defensive controls provided a comprehensive view of threats and response strategies.